Artificial Intelligence in Biomedical Informatics

Project 01: Discovering conservative patterns in the results of probing RNA-protein binding sites

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12th of March, 2026

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The objectives of the first cycle of laboratory classes are as follows:

1. To familiarize oneself with the provided datasets comprising the results of experimental probing of transcripts by the fSHAPE method in the context of two RNA-binding proteins, namely HNRNPC (*Heterogeneous Nuclear Ribonucleoprotein C*) and HNRNPA2B1 (*Heterogeneous Nuclear Ribonucleoprotein A2/B1*).
2. To conduct a cluster analysis of a set of selected motifs satisfying the assumed binding site characteristics for the protein under consideration to uncover unknown conservative patterns in the data representing potentially promising RNA-protein binding sites.

1 Introduction

1.1 Motivation and biological nature of the problem.

RNA-binding proteins (RBPs) play a key role in the process of post-transcriptional regulation. Understanding the mechanisms governing the process of protein and RNA binding and being able to locate potential promising binding sites is crucial for understanding the process of regulating gene expression. In the human genome, we distinguish hundreds of proteins that form complexes of this kind. Unfortunately, only for a small percentage of them is available the detailed knowledge of the intermolecular interactions (i.e., hydrogen bonding) occurring at the binding site between the protein and a specific motif in the RNA structure. In the vast majority of cases, current knowledge of both the sequence and secondary structure of the interacting molecules is not sufficient to accurately predict the potential binding sites of proteins of this kind within various known RNA transcripts [1,2].

As an example of a well-studied complex of this type, it is worth mentioning here a protein (IRP) that binds a specific *Iron Response Element* (IRE) motif – i.e., hairpin - located in the 5' untranslated region (so-called 5'-UTR) of the FTL transcript (mRNA). The purpose of this

interaction is to regulate the translation of the FTL gene. An example visualization of the actual RNA-protein binding is presented in Figure 1A.

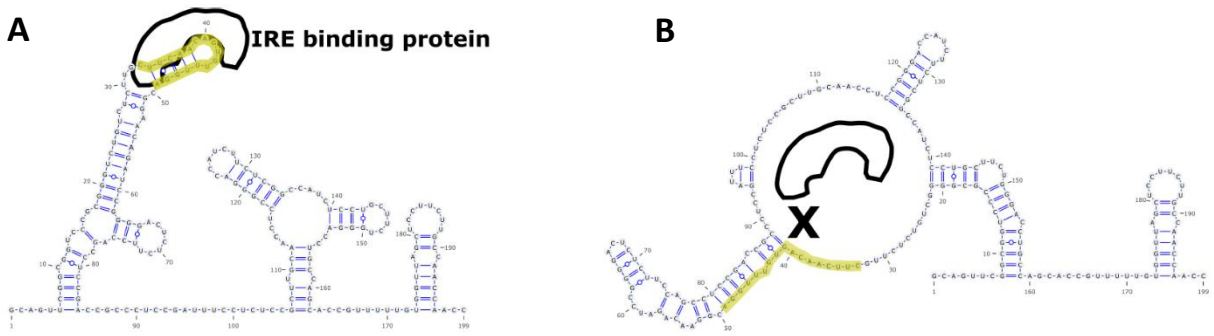

Figure 1. Schematic visualization of the protein binding site (IRP) with a hairpin located in the 5'-UTR region of the FTL transcript (source: [1,3]).

A single-nucleotide polymorphism (SNP) in the 5'-UTR of the ferritin light chain (FTL) gene causes hyperferritinemia by disrupting the translation process through a conformational change in the binding site that prevents the *Iron Regulatory Protein* (IRP) from properly interacting with the FTL 5'-UTR mRNA [5]. An example visualization of the conformation preventing RNA-protein binding is presented in Figure 1B.

1.2 Methods for experimental probing of RNA-protein binding sites.

The methods presented below make it possible to study and, in the long run, understand the specific interactions occurring between ribonucleic acids and proteins that form complexes.

1.2.1 Characteristics of the SHAPE method.

SHAPE structure probing can determine the propensity of each base in a transcript to pair with RNA. Low reactivity values (<0.4) refer to nucleotides paired (i.e., bound by hydrogen bonds) with other residues of the transcript or amino acids of the protein with which they form a complex. In contrast, high reactivity values (>0.85) indicate unpaired nucleotides that are part of single strands or loops. Within the framework of the analyzed method, three different experiments are considered (presented in Figure 2). In the first one, material in the cell (in vivo) including the transcript, its associated protein, and a specific RNA probing reagent is

sequenced. In the last one, only the transcript extracted from the cell (in vitro) supplemented with the RNA probing reagent is sequenced. Finally, the second experiment, where the material reexamined in the cell includes the transcript and its associated protein, allows the determination of baseline nucleotide reactivity values, which are then used to normalize the results obtained in the previous two experiments. The reactivity of nucleotides in the cell (in vivo) decreases noticeably for those nucleotides among the considered ones that interact (form hydrogen bonds) with the amino acids of the protein.

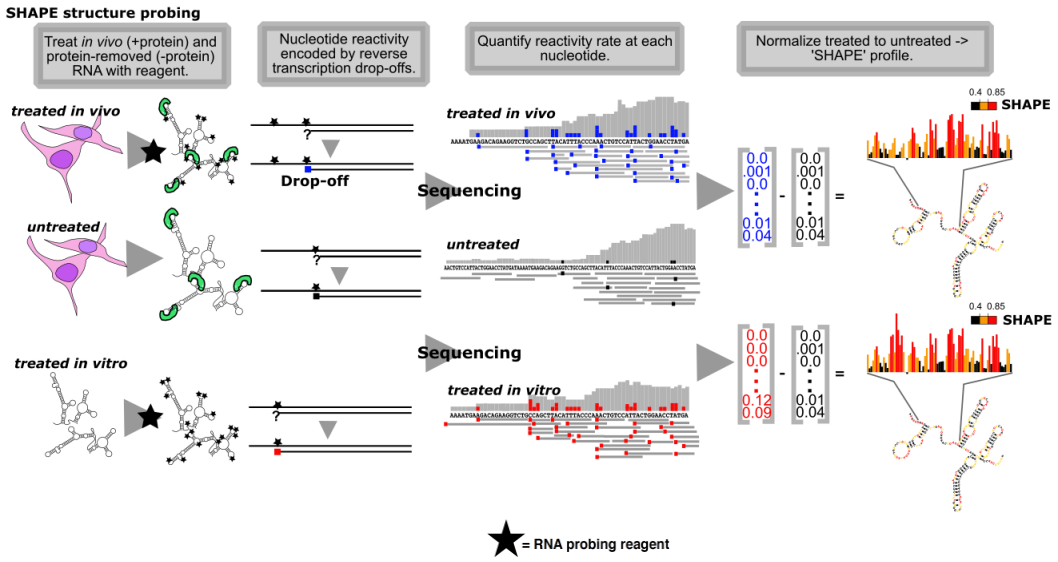

Figure 2. Concept of probing 3D RNA structures using the SHAPE method (source: [1,3]).

1.2.2 Characteristics of the fSHAPE (*footprinting SHAPE*) method.

Structure probing by fSHAPE identifies nucleotides in the transcript that interact with the protein. Structural analyses indicate that fSHAPE accurately detects nitrogenous bases in the transcript that form hydrogen bonds with the protein's amino acids. For this method, two sequencing experiments are carried out (presented in Figure 3), where the material is examined in the cell (in vivo) and outside the cell (in vitro), respectively. In both cases, a specific RNA probing reagent is used. In the case of a control experiment (in vitro), conducted to normalize nucleotide reactivity values, an RNA transcript isolated from the cell is tested. High nucleotide reactivity values (> 1.0) reported by fSHAPE clearly indicate RNA-protein binding sites.

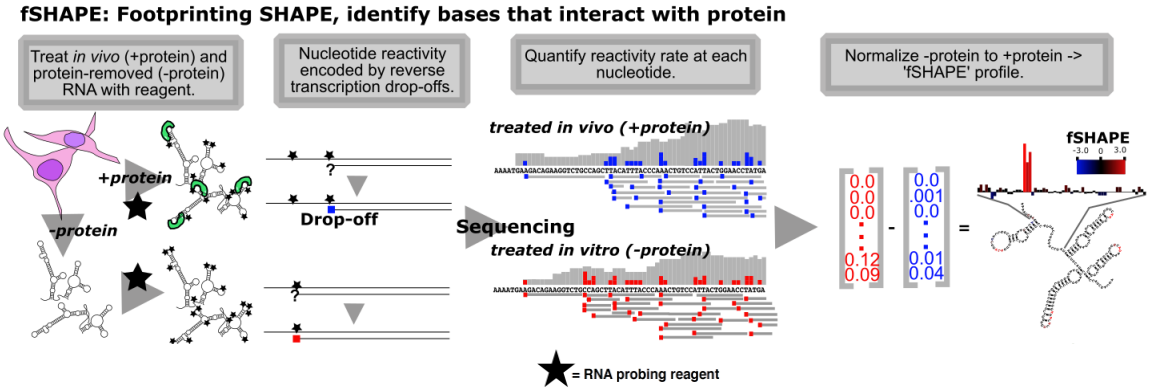

Figure 3. Concept of probing 3D RNA structures by fSHAPE method (source: [1,3]).

The fSHAPE method assumes that protein binding does not drastically change the secondary structure of the RNA, so the changes that are observed in the protein-free material are clearly due to lost interactions with the protein.

1.3 How to identify RNA-binding proteins?

The fSHAPE data profile determined for a transcript unambiguously indicates the RNA-protein binding site. For example (Figure 4), the nucleotides, and in particular their nitrogenous bases, included in the apical loop and bulge, formed within the IRE of the FTL transcript, form interactions with the *Iron* metabolism *Regulatory Protein* (IRP1) which is easily observed in the form of their high reactivity values obtained from fSHAPE probing of RNA.

Structural studies have confirmed that if we have independent transcripts and they are known to bind the same protein then experimentally determined fSHAPE data profiles for the binding sites identified within them (typically covering about 10 nucleotides) will have high correlation (> 0.8). Example fSHAPE data profiles for two transcripts binding the same iron metabolism regulatory protein are presented in Figure 5. Therefore, it is possible to discover potentially promising binding sites of known regulatory proteins for transcripts for which they have not yet been experimentally confirmed by searching for a conservative pattern in the sequence (e.g., [CG]NNNNNCAG[AT]G, where N represents any nucleotide) and verifying the relatively high correlation of fSHAPE data profiles between a known and a potentially promising binding site.

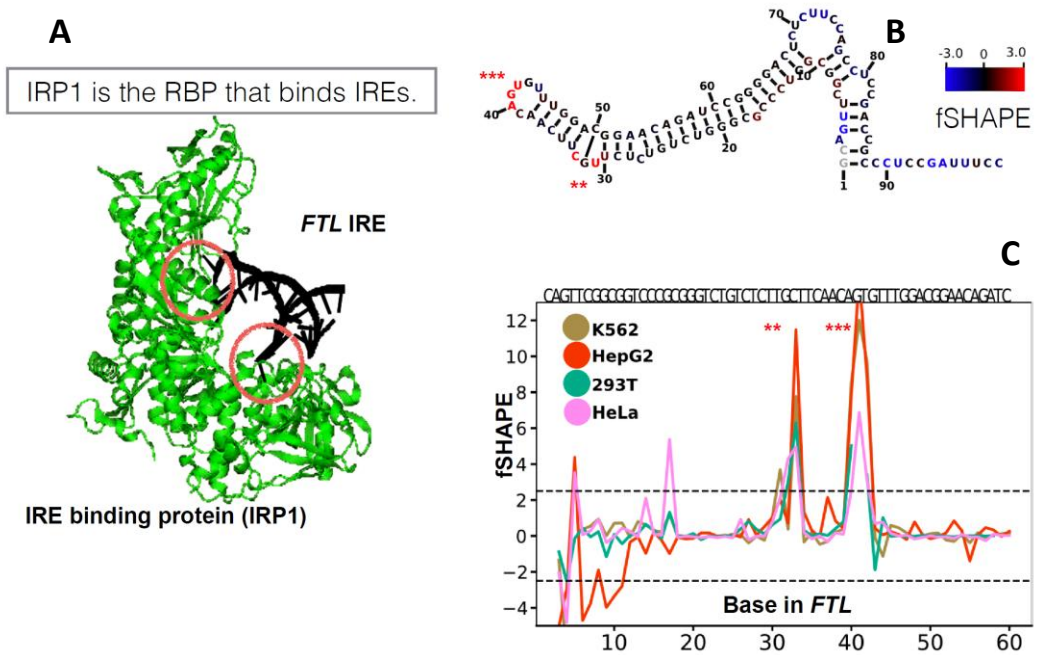

*Figure 4. Characterization of the binding site of an example RNA-protein complex. A: Visualization of the 3D structures of the protein (marked in green) and RNA fragment (marked in black), respectively, forming the complex. B: Visualization of the secondary structure of the RNA fragment, where red color indicates the nucleotides (** - bulge, *** - apical loop) whose nitrogenous bases interact with the protein. C: Diagram showing the result of probing an RNA structure fragment using the fSHAPE method (source: [1,3]).*

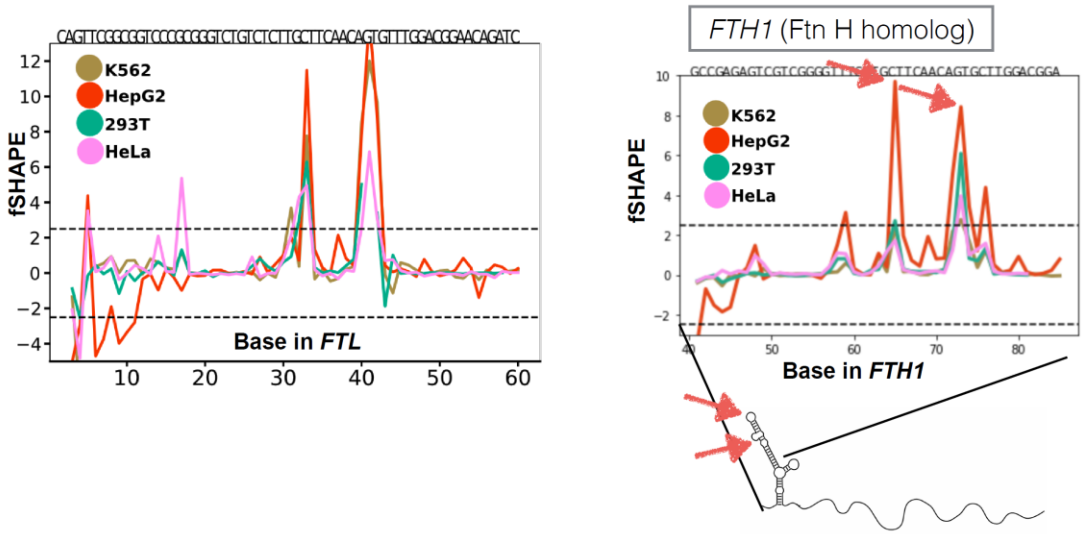

Figure 5. Profiles of fSHAPE data for fragments of FTL and FTH1 transcripts, respectively, that interact with proteins that regulate iron metabolism (source: [1,3]).

Moreover, having a set of fSHAPE datasets for transcripts known to bind a protein, but we do not know exactly which protein is bound in fact, machine learning methods can be successfully used to discover promising consensus motifs between 6 and 14 nucleotides in length satisfying assumed binding site characteristics based on the reactivity values obtained by the fSHAPE method (i.e., the reactivity value for at least one residue within the motif must exceed a value of 1.0), which can then be verified in the context of experimentally confirmed RNA-protein binding sites for which we have fSHAPE data profiles available.

1.4 Methods for discovering patterns in multiple time-series data.

The dataset obtained from probing a single transcript using the fSHAPE method is often treated as a time-series, since it includes reactivity values assigned to consecutive nucleotides included in the RNA structure under analysis. Therefore, the exploration of multiple time-series (because we usually consider many potential transcripts) should allow us to discover promising consensus/conservative motifs in them, which, after experimental confirmation, may turn out to be actual, hitherto unknown RNA-protein binding sites.

Motif discovery aims to learn new, valuable knowledge from existing data. By motif, we mean a repetitive pattern identified in the analyzed time-series without having information about its location or shape. In the case of time-series, usually rules or specific events in the analyzed signal are discovered, which often provide information useful in modeling or data analysis. Problems of this kind are considered in various fields, such as telecommunications, medicine, IoT, for example [4]. The main goal is to discover consensus/conservative motifs in the data characterized by the smallest distance occurring between them.

1.5 How to compare motives?

To determine the distance between the fSHAPE data profiles of the motifs being compared with each other, use the *z-normalized Euclidean distance* (*znEd*) measure. If the value of this measure for a specific pair of motifs does not exceed the value of 2.5 then they can be considered promising and then ranked by their sequence similarity. In the next step, the *sequence similarity factor* (*ssf*) is determined, which is the total of partial values, determined for each considered nucleotide in the motif assuming: 2 points for identically mapped nucleotides,

i.e., A<->A, G<->G, C<->C, U<->U, T<->T, 1 point for mapping in terms of purines (i.e., A<->G, G<->A) as well as pyrimidines (i.e., C<->U, U<->C, C<->T, T<->C), respectively, and 0 points otherwise, normalized by the motif length. The higher the coefficient value, the greater the similarity between motifs. For example (N stands for any nucleotide):

NTTTTN

TGATTT

Sequence similarity factor = (2 + 0 + 0 + 2 + 2 + 2)/6 = 8/6 = 1.33(3). If we want to use a single value to describe the general similarity then we can use the following formula *aS = 10 * znEd - ssf*. Of course, the smaller the value determined for *aS* the better.

1.6 Useful libraries.

MatrixProfile: https://github.com/matrix-profile-foundation/matrixprofile

STUMPY: https://github.com/TDAmeritrade/stumpy

Scikit-learn: https://github.com/scikit-learn/scikit-learn

2 Data

The datasets provided include the results of experimental fSHAPE transcript probing in the context of two RNA-binding proteins, namely HNRNPC (*Heterogeneous Nuclear Ribonucleoprotein C*) and HNRNPA2B1 (*Heterogeneous Nuclear Ribonucleoprotein A2/B1*). Within each protein under consideration, three files (one text file and two archives) are provided:

- *[hnrnpc|hnrnpa2b1]_expected_pattern* - a text file containing the expected RNA-protein binding site motif determined for a specific complex on the basis of experimentally determined spatial structure (the average number of hydrogen bonds for nucleotides considered within the protein binding site correlates with their high reactivity determined by the fSHAPE method). The file contains the data in tabular form (without a header) covering the following columns (separated by tabs), namely the fSHAPE value and the name of the base (A/C/T/G).

- *[hnrnpc|hnrnpa2b1]_binding_sites_fshape.zip* - an archive consisting of a set of short fragments of transcripts for discovering repetitive consensus/conservative patterns that may represent previously unknown, potentially promising binding sites to the protein under consideration. Each file contains data in tabular form (without a header) covering the following columns (separated by tabs), namely the fSHAPE value, the name of the base (A/C/T/G), and optionally the SHAPE value.
- *[hnrnpc|hnrnpa2b1]_ search _fshape.zip* - an archive containing a set of transcripts that will be searched using the conservative motif(s) discovered in the previous step to identify motifs with similar fSHAPE (*z-normalized Euclidean distance* should not exceed a value of 2.5) data profiles and a high sequence similarity factor.

All files should contain at least one long, continuous RNA strand fragment encompassing specific fSHAPE values. Files may also contain long strings of indeterminate (*NaN*) values, within both the SHAPE and fSHAPE columns, that should be dealt somehow with when identifying promising motifs.

3 Exercise

The exercise includes the following steps:

1. Select the protein (HNRNPC/HNRNPA2B1) analyzed during the experiments, which uniquely identifies the data set used.
2. Familiarize oneself with the expected motif, extracted from the experimentally determined 3D structure of the complex, formed by the analyzed RNA-binding protein, in order to identify the number of nucleotides (*len*) included in it. Let's assume that our expected motif is as follows TTTT, then *len* = 4.
3. Determine the length of potentially promising motifs *w = {len, len + 1, len + 2}*. Suppose our expected motif is TTTT (*len = 4*), then *len + 1 = {NTTTT, TTTTN}*, and *len + 2 = {NTTTTN}*, where N is any nucleotide.
4. Extract all promising continuous motifs (represented by the corresponding fSHAPE data profiles) of length *w = len ... len + 2* satisfying the assumed binding site characteristics, based on the reactivity values obtained by the fSHAPE method (i.e., reactivity value for

at least one nucleotide within the motif must exceed a value of 1.0), from among short fragments of transcripts contained in the archive *identifier_selected_protein_binding_sites_fshape.zip*, which are likely to contain previously undiscovered RNA binding sites to the protein under consideration.

5. Perform cluster analysis using at least two methods (e.g., *KMEANS++*, *DBSCAN*, etc.) for the set of motifs extracted in the previous step for each motif length (*w*) analyzed independently.
6. Determine a promising consensus motif(s) (e.g., using the *STUMPY* library) based on the fSHAPE data profiles contained in the three most abundant clusters identified in the previous step (the minimum power of the cluster for which the consensus motif must be determined should not be less than 3).
7. Perform a transcripts search (e.g., using the *STUMPY* library), stored in the archive *identifier_selected_protein_ search _fshape.zip*, based on the fSHAPE data profiles extracted for the promising consensus motifs identified in the previous step and the expected motif. Compile the results in tabular form, where each motif will be described by its sequence, ranges of nucleotide numbers, the name of the transcript file in which it was identified, and the values of the following measures, namely *znEd*, *ssf*, *αS* determined in the context of the expected motif. The records should be ordered in non-decreasing order according to the last column (*αS*).

After completing the exercise, each group prepares a short presentation (~3 slides) showing the proposed solution with justification and the results obtained.

Please complete the exercise before April 8th.

4 Literature

1. Corley, M., Antczak, M., Zok, T., Elhajjajy, S., Bernetti, M., & Adler J., (2021) "Predicting hnRNPA2/B1 binding sites from fSHAPE data", RBP Footprint Challenge @ RNA Society 2021 Meeting, 04.06.2021
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footprints from structural data. *RNA*, 31(8), 1103-1124, https://doi.org/10.1261/rna.080215.124

3. Corley, M., Burns, M. C., & Yeo, G. W. (2020). How RNA-binding proteins interact with RNA: molecules and mechanisms. *Molecular Cell*, 78(1), 9-29, https://doi.org/10.1016/j.molcel.2020.03.011
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5. Torkamani, S., & Lohweg, V. (2017). Survey on time series motif discovery. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 7(2), e1199, https://doi.org/10.1002/widm.1199
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